

# 2026 BEDAŞ Predictive Maintenance

December 2025

## 1 Introduction

Urban street lighting systems constitute a critical component of municipal infrastructure, operating continuously under highly structured temporal schedules. Under normal conditions, their electricity consumption follows a predictable daily pattern: stable load during nighttime hours when luminaires are active, near-zero consumption during daylight, and sharp transitions at the evening switch-on and morning switch-off periods. This regularity makes street lighting networks particularly well-suited to data-driven anomaly detection: deviations from the expected consumption profile are inherently meaningful and can serve as early indicators of equipment malfunction, relay failure, or communication irregularities in the remote management system.

For large-scale operators such as BEDAŞ, which manages one of the largest urban lighting networks in Turkey, maintaining operational reliability across thousands of distributed field units is a significant challenge. Manual inspection of consumption records is neither scalable nor timely; by the time a fault is identified through routine maintenance, service interruptions may already have occurred. Automated anomaly detection offers a more systematic alternative: by continuously monitoring consumption data against learned baselines, deviations can be flagged at the moment they occur, enabling targeted intervention before failures escalate.

This study analyzes 14 months of hourly electricity consumption data obtained from the BEDAŞ Automated Meter Reading System (OSOS), covering the period from January 2025 to February 2026. The primary objective is to identify consumption behaviors that deviate significantly from established temporal patterns and to characterize the distribution of these deviations across time. Four independent anomaly detection approaches are applied to the same dataset, enabling a comparative evaluation of their respective strengths and limitations in the context of urban lighting infrastructure.

## 2 Methodology

The dataset used in this study consists of hourly active power withdrawal measurements recorded by the OSOS system for the BEDAŞ urban lighting network.

Data were provided in monthly Excel files, each containing hourly consumption readings for every calendar day of the corresponding month. The observation window spans January 2025 to February 2026, yielding 14 months of measurements. Only records labeled as power withdrawal were retained for analysis, as these reflect the actual energy consumed by the lighting network rather than bidirectional metering artifacts.

Four anomaly detection methods were independently applied to the dataset. Each method addresses the detection problem from a distinct modeling perspective, with preprocessing and feature engineering steps tailored to the requirements of the respective algorithm. The methods and their results are described in the following subsections.

## 2.1 Isolation Forest-Based Anomaly Detection

### 2.1.1 Data Acquisition and Preprocessing

Monthly electricity consumption datasets provided in Excel format by BEDAŞ’s Automated Meter Reading System (OSOS) were loaded into a Python environment and consolidated into a single longitudinal dataset spanning 14 months, from January 2025 to February 2026. Since the raw files contained both feed and withdrawal measurements, only records labeled as *Çekiş* (active power withdrawal) were retained, ensuring the analysis reflects actual energy consumed by the lighting network rather than bidirectional meter readings.

The dataset was subsequently restructured from a wide format—where each column represented a calendar day—into a long format in which each row corresponds to a single hourly observation. Proper timestamp variables were constructed by combining year, month, day, and hour fields extracted from the source files. Missing hourly records, arising primarily from metering communication gaps, were addressed through time-based linear interpolation, which estimates absent values proportionally between adjacent observations, preserving the continuity of the time series without introducing artificial discontinuities. Following preprocessing, the final dataset comprised **9,780 hourly observations**.

### 2.1.2 Algorithm Selection

Isolation Forest was selected as the anomaly detection algorithm for this study for three principal reasons. First, the absence of verified fault records in the OSOS dataset made supervised approaches inapplicable; Isolation Forest operates in a fully unsupervised manner, requiring no labeled examples of anomalous behavior. Second, the algorithm is computationally efficient on large time-series datasets, as it constructs an ensemble of random decision trees without requiring distance or density computations. Third, it imposes no assumption on the underlying data distribution, making it well-suited to the irregular consumption patterns characteristic of street lighting systems.

The algorithm operates on the principle that anomalous observations are “*few and different*”: in a forest of randomly constructed decision trees, anoma-

lous data points reach leaf nodes in significantly fewer partitions than typical observations, yielding shorter average path lengths and correspondingly lower anomaly scores. The key hyperparameters used in this study are summarized in Table 1.

Table 1: Isolation Forest Hyperparameter Configuration

| Parameter                  | Value | Justification                                                                       |
|----------------------------|-------|-------------------------------------------------------------------------------------|
| <code>n_estimators</code>  | 200   | Ensemble stability across large dataset                                             |
| <code>contamination</code> | 0.03  | $\approx 3\%$ outlier rate estimated from visual inspection of consumption profiles |
| <code>random_state</code>  | 42    | Reproducibility                                                                     |

### 2.1.3 Feature Engineering

Raw consumption values alone are insufficient for meaningful anomaly detection in this context: a value of 7 kWh, for example, is entirely normal at 02:00 when lights are expected to be on, yet represents a significant deviation at 14:00 when the network should be drawing near-zero power. To enable the model to understand this temporal context, several additional features were derived from the timestamp index, as described in Table 2.

Table 2: Engineered Features Used in Model Training

| Feature               | Description                                | Rationale                                                                                  |
|-----------------------|--------------------------------------------|--------------------------------------------------------------------------------------------|
| Consumption           | Hourly active power draw (kWh)             | Primary anomaly signal                                                                     |
| <code>hour_sin</code> | $\sin\left(\frac{2\pi \cdot h}{24}\right)$ | Circular encoding: preserves temporal                                                      |
| <code>hour_cos</code> | $\cos\left(\frac{2\pi \cdot h}{24}\right)$ | continuity between hours 23:00 and 00:00                                                   |
| Day of week           | Integer 0–6 (Mon–Sun)                      | Captures weekly schedule variation                                                         |
| <code>lag24</code>    | Consumption 24 hours prior                 | Same-hour previous-day baseline; flags abrupt deviations from expected repetitive behavior |

Sinusoidal encoding was applied to the hour variable because a linear representation treats hours 23 and 0 as maximally distant, whereas operationally they are contiguous. The `lag24` feature exploits the highly repetitive daily schedule of street lighting systems: since normal consumption at any given hour closely mirrors the same hour on the preceding day, a significant departure from `lag24` constitutes a direct anomaly signal. All features were standardized using `StandardScaler` prior to model fitting.

#### 2.1.4 Two-Phase Detection Pipeline

The analysis was conducted in two sequential phases, each addressing a distinct objective. The motivation for this two-phase design is straightforward: applying Isolation Forest to the entire dataset at once offers a useful exploratory overview, but provides no evidence of how well the model generalizes to data it has never seen. In a real predictive maintenance context, the model must be capable of identifying future faults—not just patterns in historical data. The second phase was designed specifically to answer this question.

**Phase 1 — Global Anomaly Detection.** In the first phase, Isolation Forest was applied across the full 14-month dataset. The objective was exploratory: to examine the overall distribution of anomalies, determine whether they clustered during specific periods, and establish a system-wide baseline. Each observation was classified as either normal or anomalous, and results were aggregated by month, day, hour, and meteorological season. This phase yielded **294 detected anomalies**, with the seasonal breakdown shown in Table 3.

Table 3: Seasonal Distribution of Anomalies — Phase 1 (Global)

| Season                 | Anomalies  | Normal       | Anomaly Rate |
|------------------------|------------|--------------|--------------|
| Winter (Dec, Jan, Feb) | 135        | 3,052        | 4.2%         |
| Summer (Jun, Jul, Aug) | 72         | 2,136        | 3.3%         |
| Spring (Mar, Apr, May) | 49         | 2,152        | 2.2%         |
| Autumn (Sep, Oct, Nov) | 38         | 2,146        | 1.7%         |
| <b>Total</b>           | <b>294</b> | <b>9,486</b> | <b>3.0%</b>  |

Winter accounted for 45.9% of all flagged observations. This finding aligns with expectations: lower temperatures and extended lighting hours increase the load on network infrastructure, making faults more likely during this period.

**Phase 2 — Train-Test Based Evaluation.** To measure how well the model performs on unseen data, the dataset was split chronologically. All observations prior to December 2025 formed the training set, while the remainder constituted the test set. The `StandardScaler` normalization was computed on the training set alone and applied to the test set without refitting, preventing any future information from influencing the training process. This setup simulates a realistic deployment scenario: the model learns from historical data and is then applied to a future period it has never encountered. The split statistics are summarized in Table 4.

Table 4: Train-Test Split Statistics — Phase 2

| Set                            | Observations | Anomalies Detected | Anomaly Rate |
|--------------------------------|--------------|--------------------|--------------|
| Training (Jan 2025 – Nov 2025) | 7,979        | —                  | —            |
| Test (Dec 2025 – Feb 2026)     | 1,777        | 162                | 9.1%         |

The test-period anomaly rate of 9.1% substantially exceeds the 3% contamination prior specified during training. This should not be interpreted as model miscalibration. Rather, it is a direct consequence of the seasonal pattern identified in Phase 1: the model learned normality from predominantly non-winter data, and correctly identified the December–February test period as one of heightened operational deviation.

## 2.2 GAF-Based Pattern Analysis

The Gramian Angular Field (GAF) approach was employed as a pattern-based anomaly detection method to analyze daily electricity consumption profiles at the structural level. While point-based methods focus on individual observations, the GAF representation makes it possible to evaluate the full 24-hour operating profile of each day as a coherent pattern. This was particularly useful in the present study, since two days with similar total electricity consumption may still differ substantially in terms of nighttime intensity, transition timing, and internal temporal organization.

### Daily Input Structure:

The GAF-based analysis was performed on daily electricity consumption vectors derived from the OSOS dataset covering the period from 2025-01-01 to 2026-02-13. For each of the 409 days in the dataset, hourly electricity consumption values from 00:00 to 23:00 were arranged into a 24-dimensional vector. These daily vectors constituted the basic input for the image transformation and anomaly scoring stages.

### Missing Data Handling:

Before applying the GAF transformation, the dataset was preprocessed in order to distinguish true missing values from genuine zero consumption. This distinction was essential, because treating missing hourly observations as zeros would artificially distort the daily load profile and could lead to false anomaly signals.

- **Presence mask:** For each day, a binary mask was defined, where 1 indicates that a measurement was recorded and 0 indicates a missing value.
- **Missing ratio:** The proportion of missing hourly observations was computed as

$$\text{missing\_ratio} = \frac{n_{\text{missing\_hours}}}{24}$$

- **Complete days:** Days with 0 missing hours were labeled as `complete`.
- **Partial missing days:** Days with 1 to 6 missing hours were labeled as `partial_missing`.
- **Low-quality days:** Days with more than 6 missing hours were labeled as `data_quality_low` and excluded from the final anomaly ranking.

In the final dataset, 399 days were complete, 8 days were partial missing, and 2 days were classified as low quality. In total, 36 hourly measurements were missing out of 9816 hourly observations, corresponding to 0.37% missingness. The 2 low-quality days were excluded from scoring, leaving 407 days for the final GAF-based analysis.

**Imputation Strategy:**

Since the GAF transformation requires a complete 24-dimensional vector, missing hourly values in partial-missing days were imputed using the monthly hourly median. In other words, a missing value at a given hour was replaced by the median value of that same hour within the same calendar month. This imputation was applied only to enable image generation and preserve the expected hourly structure of the daily profile. Across the dataset, 36 hourly values across 10 days were imputed for GAF input.

**GAF Transformation:**

After preprocessing, each daily 24-hour vector was transformed into a  $24 \times 24$  Gramian Angular Summation Field (GASF) image. The transformation was carried out in three stages:

- **Normalization:** The daily consumption vector was normalized to the interval  $[-1, 1]$ .
- **Angular encoding:** The normalized values were mapped into angular form using the arccosine function.
- **Matrix construction:** The GAF image was computed according to

$$GAF(i, j) = \cos(\phi_i + \phi_j)$$

This transformation converts the one-dimensional daily time series into a two-dimensional image representation that captures pairwise temporal relationships between all hours of the day. As a result, the method preserves not only consumption magnitude but also the structural arrangement of the daily operating pattern.

**Structural Interpretation of GAF Images:**

The GAF representation made it possible to distinguish normal and abnormal daily operating structures more clearly than raw time-series plots alone.

- **Normal days:** Normal daily profiles exhibited a clear symmetric block structure. The upper-left and lower-right regions, corresponding to night-night and evening-evening relationships, showed strong positive values. Cross-regions between high nighttime consumption and near-zero daytime consumption showed strong negative values.
- **Anomalous days:** Abnormal daily profiles showed varying degrees of structural distortion, including weaker block contrast, shifted color boundaries, asymmetric regions, and irregular off-diagonal patterns.

These structural changes indicate deviations from the dominant street-lighting behavior, even in cases where the raw daily profile may not appear dramatically different at first sight.

#### **Anomaly Scoring:**

To quantify structural deviation, an unsupervised autoencoder model was applied to the generated GAF images. The model architecture used in the analysis was 576-64-16-64-576. The model was trained on GAF images derived from complete, non-zero consumption days, and reconstruction error was used as the raw anomaly score. The anomaly score was defined as the mean squared reconstruction error, normalized to the interval  $[0, 1]$ . In this framework, regular daily patterns are reconstructed more accurately, whereas structurally unusual days yield higher reconstruction error.

#### **Mask-Aware Final Score:**

Because some retained days still contained imputed values, the final anomaly ranking was adjusted using a mask-aware weighting scheme. The final score was computed as

$$\text{mask\_aware\_score} = \text{raw\_reconstruction\_error} \times \left( \frac{n_{\text{present}}}{24} \right)$$

where  $n_{\text{present}}$  denotes the number of actually recorded hourly values for that day. This weighting ensures that days with a small number of missing hours remain analyzable, while days with more incomplete records receive proportionally reduced scores. In this way, the method reduces the risk of inflating anomaly scores purely because of missing data.

#### **Full-Period Findings:**

After data-quality filtering, 407 days were included in the final GAF-based analysis. The top-ranked anomalous days were mainly characterized by reduced nighttime consumption amplitude and visible structural distortion in their GAF images. According to the ranked results, the highest anomaly scores were observed on 2025-01-04, 2025-01-06, 2025-03-17, 2025-02-03, and 2025-03-12. In addition, a persistent structural change was identified from mid-December 2025 onward, where NightMean consumption decreased from approximately 6.8 kWh to 5.2 kWh, corresponding to a reduction of 24.6%. Unlike isolated anomalies observed in earlier months, this late-period change persisted for several weeks and therefore indicated a more sustained system-level deviation.

#### **Methodological Interpretation:**

The GAF-based analysis showed that anomaly candidates were primarily associated with deviations in the internal structure of daily electricity consumption behavior. In particular, the most prominent anomalous days were characterized by reduced nighttime intensity, altered transition regions, weaker block contrast, or fragmented image structure. Since maintenance logs and direct field validation were not available, these results should be interpreted as anomaly indicators rather than definitive fault labels. Accordingly, the GAF-based method is best understood as a decision-support tool for identifying suspicious daily profiles that may require further technical investigation.

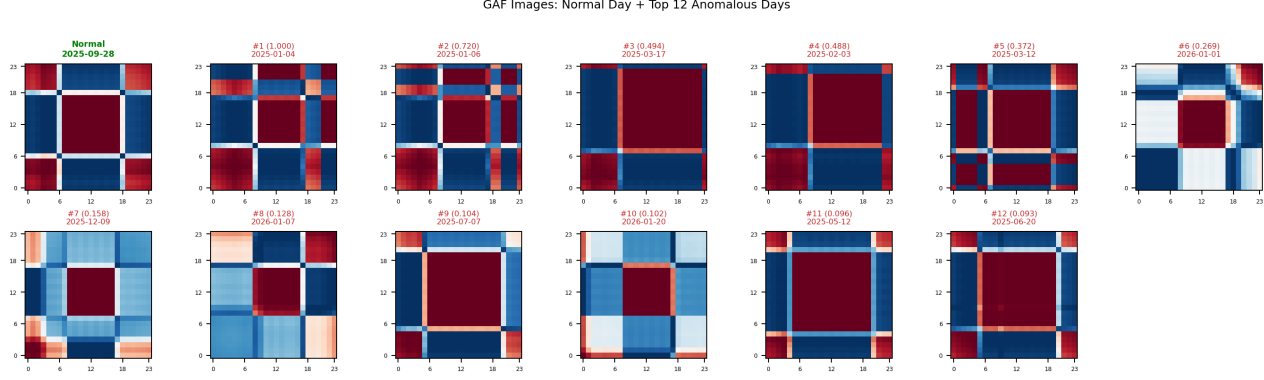


Figure 1: Representative GAF images of one normal day and the top anomalous days.

### 2.3 Data and Feature Engineering for Supervised and Anomaly Detection Models

The supervised and anomaly detection models described in this section were developed using a merged dataset that combines energy consumption records with three-phase electrical measurements from the same meter. The energy dataset contains 9,780 hourly records and the electrical dataset contains 9,456 hourly records, covering the same 14-month period. The two sources were merged on their shared timestamp, and all subsequent analyses were conducted on the resulting unified dataset.

We defined the prediction target as a binary label indicating whether a failure will start within the next 24 hours ( $y_t^{24h}$ ), yielding 547 positive samples (5.59% of all hours). This horizon was chosen because it provides a practical window for dispatching maintenance crews while retaining enough positive samples for supervised learning.

Feature engineering produced two feature sets. The **energy feature set** (16 features) includes rolling mean and standard deviation over 3, 6, 12, and 24-hour windows, hourly and daily differencing, and temporal indicators (hour of day, day of week, month, weekend/night flags). The **merged feature set** (53+ features) extends this with phase current imbalance ( $\sigma_I = \text{std}(I_R, I_S, I_T)$ ), voltage imbalance, average current/voltage/power factor across phases, apparent and active power, energy-to-current ratio, and rolling standard deviations of electrical parameters over 6, 12, and 24-hour windows. All features were standardized using z-score normalization.

### 2.4 Supervised Failure Prediction (Gradient Boosting)

Supervised failure prediction was selected because the availability of 28 labeled failure events—however limited—allows the model to learn discriminative pat-



terns that distinguish pre-failure hours from normal operation. Unsupervised methods can flag unusual patterns, but they cannot distinguish between “unusual but harmless” and “unusual because a failure is coming.” Supervised learning addresses this by directly optimizing for failure prediction accuracy.

#### 2.4.1 Why Gradient Boosting?

We selected Gradient Boosting Decision Trees (GBDT) as the classifier for three reasons. First, GBDT models consistently achieve state-of-the-art performance on tabular datasets [?], particularly when the number of samples is small to moderate—a critical consideration given our  $\sim 9,500$  training rows. Second, GBDT provides feature importance scores that allow us to verify whether the model relies on physically meaningful signals (e.g., current imbalance) rather than spurious correlations. Third, the model’s sequential tree construction naturally handles feature interactions (e.g., the combination of low voltage *and* high current imbalance) without requiring explicit interaction terms.

The model was trained with 300 estimators, a maximum depth of 5, a learning rate of 0.05, row subsampling of 80%, and a minimum of 10 samples per leaf. All evaluation used a time-based 80/20 train/test split (no shuffling) to prevent temporal data leakage.

#### 2.4.2 Why Merged Features?

Before addressing class imbalance, we first tested whether electrical measurements add predictive value beyond energy consumption. We trained identical GradientBoosting models on the energy-only feature set (Model A, 16 features) and the merged feature set (Model B, 53 features) using five-fold time-series cross-validation.

Model B outperformed Model A on every metric: F1 improved from 0.0435 to 0.0555 and ROC-AUC from 0.4285 to 0.5063. While both scores are modest in absolute terms, the improvement is consistent across all folds and the direction is unambiguous—ROC-AUC moves from below random chance (0.43) to above the random baseline (0.51). This confirmed that electrical measurements carry independent predictive signal, and all subsequent experiments used the merged feature set.

#### 2.4.3 Why Threshold Optimization Over Oversampling?

The extreme class imbalance (5.59% positive rate) means that a default classification threshold of 0.50 produces near-zero recall: the model almost never predicts a failure because the prior probability is so low. The standard remedy in the literature is synthetic oversampling—generating artificial positive samples to balance the training distribution. We evaluated three oversampling methods: SMOTE [?], which interpolates between existing positive samples; SMOTE combined with Tomek link removal [?], which additionally removes ambiguous borderline samples; and ADASYN [?], which focuses synthetic generation on harder-to-learn regions.

However, we also tested a simpler alternative: keeping the original training distribution but sweeping the classification threshold from 0.03 to 0.60 to find the value that maximizes F1-score. The rationale is that oversampling modifies the data the model learns from—potentially introducing artifacts in time-series data where temporal continuity matters—while threshold tuning modifies only the decision boundary after learning.

## 3 Discussion

### 3.1 Isolation Forest Findings and Interpretation

The two-phase Isolation Forest analysis applied to 14 months of hourly electricity consumption data yields three principal findings: anomaly occurrence is strongly seasonal, with winter months accounting for nearly half of all detected deviations; anomalies cluster at the two daily lighting transition windows rather than during stable operating periods; and a model trained on predominantly non-winter data produces an elevated anomaly rate when applied to a winter test period—a result that reflects a genuine seasonal shift in system behavior rather than a modeling artifact.

#### 3.1.1 Seasonal Anomaly Distribution

Figure 2 shows the monthly anomaly counts across the full observation window. January 2025 and January 2026 recorded the highest counts (38 and 39, respectively), while March 2025 produced the lowest (5). The near-identical anomaly volumes in the same calendar month one year apart—drawn from entirely independent data files—provide strong evidence that the detected pattern reflects a genuine recurring seasonal phenomenon in the physical infrastructure rather than a coincidence of the model’s decision boundary.

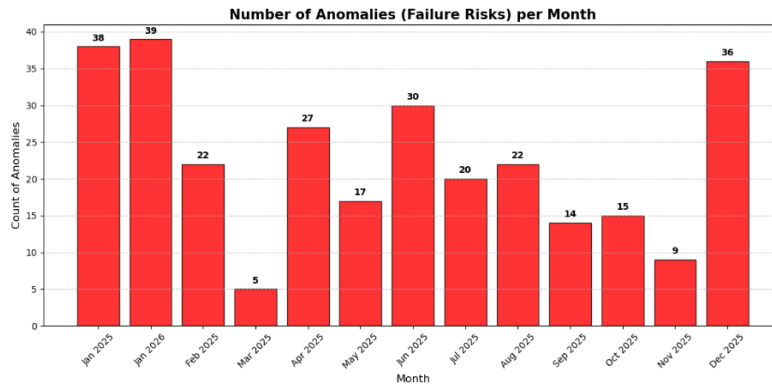


Figure 2: Number of detected anomalies per month across the 14-month observation window (Phase 1, global detection).

Winter months collectively accounted for 45.9% of all 294 flagged observations. Extended operating hours during winter increase the cumulative thermal stress on relay contacts and driver circuits, raising the probability of measurable consumption deviations. The sharp drop from February (22) to March (5) aligns with the expected seasonal load reduction and further supports the validity of the anomaly signal.

### 3.1.2 Hourly Anomaly Clusters and Operational Interpretation

The 24-hour anomaly heatmap (Figure 3) reveals a structurally consistent bimodal pattern across all 14 months: anomalies concentrate at **07:00** and **17:00–18:00**, corresponding to the morning switch-off and evening switch-on transitions of the lighting network.

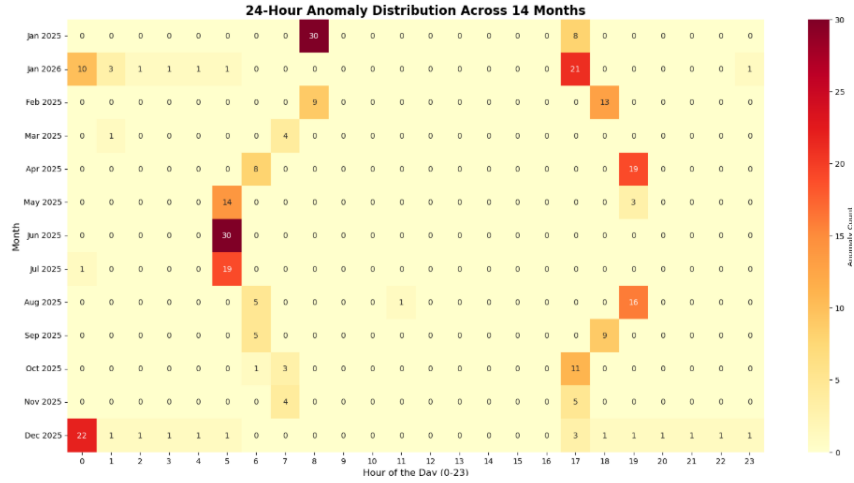


Figure 3: 24-hour anomaly distribution across 14 months (Phase 1). Anomalies concentrate at hours 07:00 and 17:00–18:00, corresponding to daily lighting transition events.

These windows are operationally significant: relay activation and deactivation cycles are the moments most susceptible to timing faults and switching irregularities. A relay that triggers late or a photocell malfunction would manifest as a consumption reading that deviates from the expected step change at these hours. The virtual absence of anomalies during midday hours (10:00–16:00) confirms that the network draws near-zero power during daylight and that the model correctly identifies this window as structurally normal. This finding has a direct maintenance implication: automated alerts triggered specifically during the 06:00–08:00 and 16:00–19:00 windows would improve fault detection efficiency while reducing unnecessary review of stable operating periods.

### 3.1.3 Anomaly Calendar and the December 31 Cluster

The day-month heatmap (Figure 4) shows that most days record zero or one anomaly, indicating broad conformity with expected consumption behavior. However, December 31, 2025 stands out with 13 anomalies—the highest single-day count in the dataset—followed by January 1, 2026 with 7.

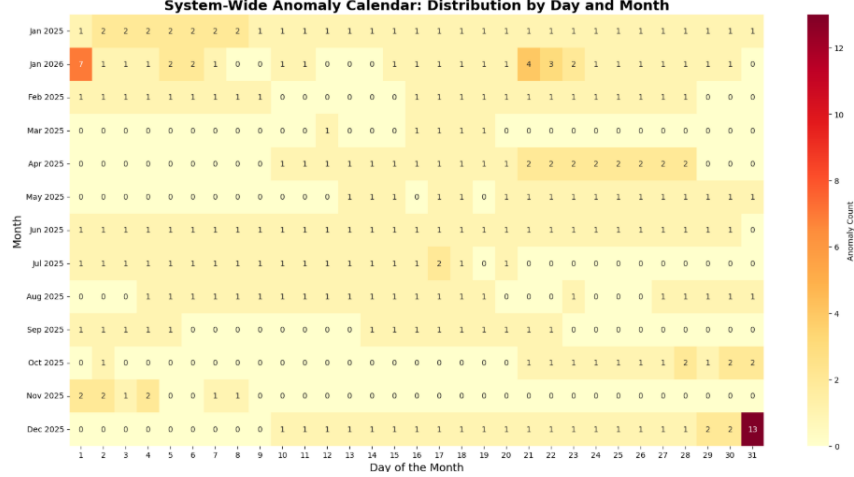


Figure 4: System-wide anomaly calendar: daily anomaly counts by month (Phase 1). The Dec 31 / Jan 1 cluster is the most pronounced localized concentration in the dataset.

This two-day cluster coincides with the New Year period, during which municipalities commonly extend lighting schedules for public events. The model correctly flags these readings as statistically anomalous relative to the learned baseline, yet they may reflect deliberate schedule changes rather than equipment faults. This illustrates a fundamental characteristic of unsupervised detection: the model cannot distinguish between a consumption deviation caused by a relay fault and one caused by an intentional but undocumented operating change. Integrating calendar event metadata into the feature set would allow such known deviations to be excluded from the anomaly score.

### 3.1.4 Phase 2 Generalization to Unseen Data

Figure 5 shows anomaly predictions on the December 2025 – February 2026 test period. Flagged observations cluster at points where the consumption signal departs from the characteristic sawtooth pattern of nightly peaks and daytime troughs, with notable concentrations around the New Year period.

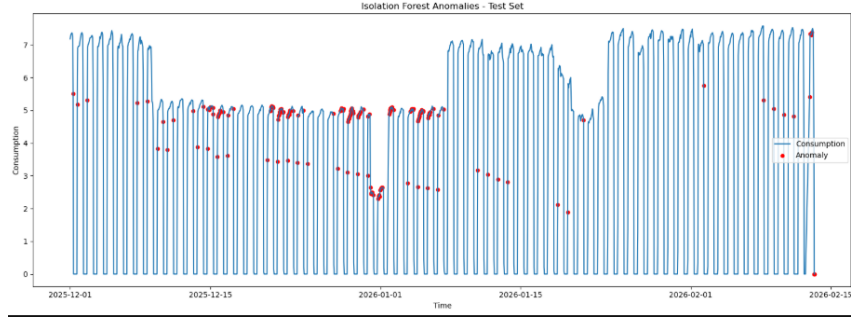


Figure 5: Isolation Forest anomaly detection on the test set (Dec 2025 – Feb 2026). Red markers indicate flagged observations.

Figure 6 confirms that the bimodal transition signature identified in Phase 1 is preserved in the test period: hours 07:00 and 17:00 again emerge as dominant anomaly windows with 28 and 29 flagged observations respectively, indicating that the model has successfully generalized the learned switching-transition pattern to unseen data.

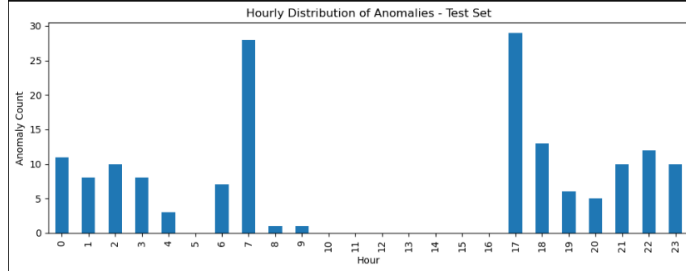


Figure 6: Hourly distribution of anomalies in the test set (Phase 2). The bimodal pattern at 07:00 and 17:00 mirrors the Phase 1 result.

### 3.1.5 The Contamination Discrepancy: Why 3% Became 9.1%

The test-period anomaly rate of 9.1% substantially exceeds the 3% contamination prior, and this discrepancy warrants explicit discussion. The `contamination` parameter does not define the expected anomaly rate in the data; it positions the decision threshold such that 3% of *training* observations are classified as anomalous. This threshold is then held fixed and applied to the test set without adjustment.

The training set (January – November 2025) contains only two winter months (January and February 2025) against nine non-winter months, so the learned normality baseline is weighted toward spring, summer, and autumn consumption behavior. When applied to a test set consisting entirely of winter months, the model encounters a distribution that is systematically shifted relative to

training: longer operating hours produce higher nighttime consumption, and extended switching cycles increase the frequency of marginal readings near the decision boundary.

Figure 7 illustrates this effect directly: December accounts for 100 of the 162 test anomalies (61.7%), January for 49 (30.2%), and February for only 13 (8.0%). This declining trajectory mirrors the gradual retreat of winter operating conditions and confirms that the elevated rate is driven by a seasonal distribution shift rather than model miscalibration. In a production deployment, a seasonally adaptive threshold—calibrated separately for winter and non-winter periods—would produce more stable and interpretable anomaly rates across the full annual cycle.

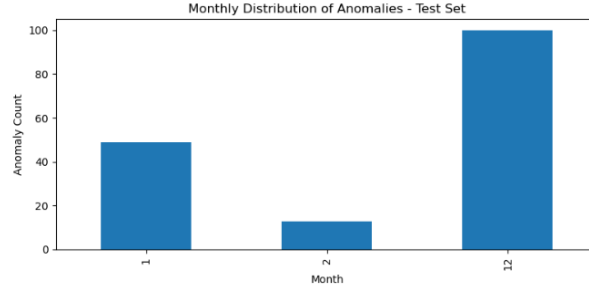


Figure 7: Monthly distribution of anomalies in the test set (Phase 2). December dominates with 100 flagged observations, reflecting the seasonal mismatch between the training distribution and the winter test period.

### 3.1.6 Practical Implications

The findings translate into three concrete recommendations for the management of the BEDAŞ lighting network. First, preventive maintenance campaigns should be front-loaded into October and November, before the onset of the high-stress winter period when anomaly rates are historically highest. Second, automated alerts should be configured to trigger during the 06:00–08:00 and 16:00–19:00 windows, where fault detection sensitivity is greatest. Third, calendar event metadata should be integrated into the anomaly pipeline to separate known schedule changes from unexpected equipment deviations, reducing false positives during public holiday periods.

### 3.1.7 Limitations

Several limitations must be acknowledged. The absence of verified fault records means that detected anomalies cannot be evaluated against confirmed infrastructure failures, and precision and recall cannot be computed without ground-truth annotations. The contamination parameter was estimated through visual inspection rather than derived from historical fault frequencies. The model assigns a single binary label to all flagged observations without distinguishing

between fault types, and the feature set does not include external covariates such as sunrise and sunset times or ambient temperature. Future work could address these gaps by incorporating verified maintenance records, extending the feature set with meteorological covariates, and applying a seasonally adaptive contamination threshold.

### 3.2 Model 2 Results: Supervised Prediction

Table 5 presents the results of the oversampling comparison. The baseline model without oversampling, when combined with threshold optimization (threshold = 0.09), achieved the highest F1-score of 0.101 with 65.9% recall—meaning it correctly identified 27 out of 41 failure-adjacent hours. This substantially outperformed all three oversampling strategies on the binary classification metric that matters most for operational deployment.

Table 5: Supervised model performance across oversampling strategies. All models use GradientBoosting with merged features. Thresholds are optimized per strategy.

| Strategy        | Threshold | Precision | Recall       | F1           | ROC-AUC      |
|-----------------|-----------|-----------|--------------|--------------|--------------|
| No oversampling | 0.09      | 5.5%      | <b>65.9%</b> | <b>0.101</b> | 0.664        |
| SMOTE           | 0.05      | 3.9%      | 43.9%        | 0.072        | 0.689        |
| SMOTE+Tomek     | 0.05      | 4.7%      | 56.1%        | 0.087        | <b>0.725</b> |
| ADASYN          | 0.05      | 5.0%      | 61.0%        | 0.092        | 0.699        |

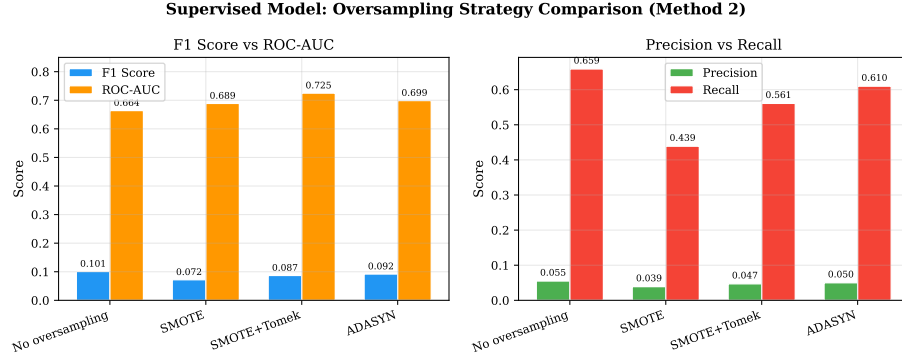


Figure 8: Left: F1 score vs ROC-AUC by strategy. The baseline achieves the best F1 through threshold tuning; SMOTE+Tomek achieves the best ROC-AUC. Right: precision–recall trade-off, showing all strategies sacrifice precision for recall.

SMOTE+Tomek achieved the best ROC-AUC (0.725), indicating that its probability outputs rank hours by failure risk more reliably than the other

strategies. However, its F1 score (0.087) was lower than the threshold-tuned baseline. This apparent contradiction arises because ROC-AUC measures the quality of the full probability ranking, while F1 measures binary classification at a single threshold. SMOTE+Tomek produces better-calibrated probabilities, but the oversampling process—which creates synthetic positive samples by interpolating between existing ones—disrupts the temporal structure of the time series, causing the model to learn slightly distorted decision boundaries.

This result has a practical implication: if the deployment goal is binary alerting (“dispatch a crew: yes/no”), threshold-tuned baseline is preferable. If the goal is risk ranking (“which meters are highest risk today?”), SMOTE+Tomek is preferable. We recommend the latter, as risk scoring tolerates imprecision better than binary alerting.

Feature importance analysis (Figure 9) confirmed that the model relies on physically interpretable signals. The top energy features were 24-hour rolling standard deviation and mean of energy consumption, which capture whether consumption stability is degrading. Among electrical features, current imbalance—the standard deviation across the three phases—ranked fifth overall, consistent with domain knowledge that asymmetric phase loading is a precursor to equipment faults. Phase voltage (R\_voltage), power factor (R\_pf), and 24-hour current differencing also contributed meaningfully. No feature that lacks physical interpretation appeared in the top ranks, providing confidence that the model captures genuine failure precursors.

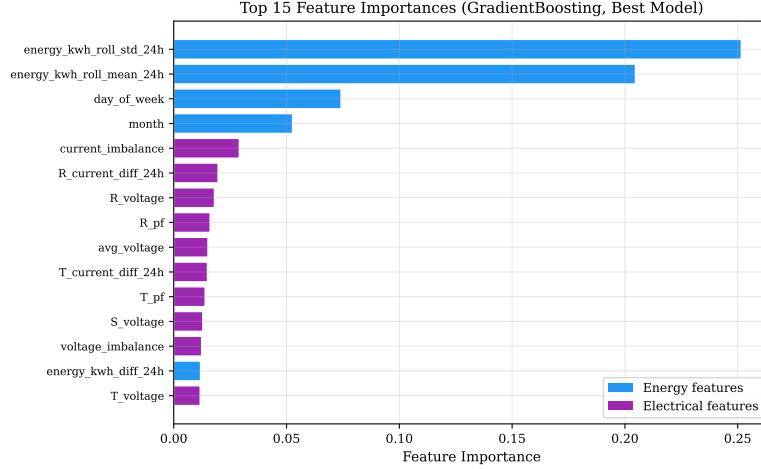


Figure 9: Top 15 feature importances from the best GradientBoosting model. Blue: energy features; purple: electrical features. Electrical features including current imbalance, phase voltage, and power factor contribute to 8 of the top 15 positions.

The comparison between energy-only and merged models (Figure 10) demon-



strates that electrical features improve every metric. The improvement in recall is the most operationally significant: the merged model detects more failure-adjacent hours because it captures failure types—such as those manifesting primarily through current imbalance—that leave no trace in energy consumption alone.

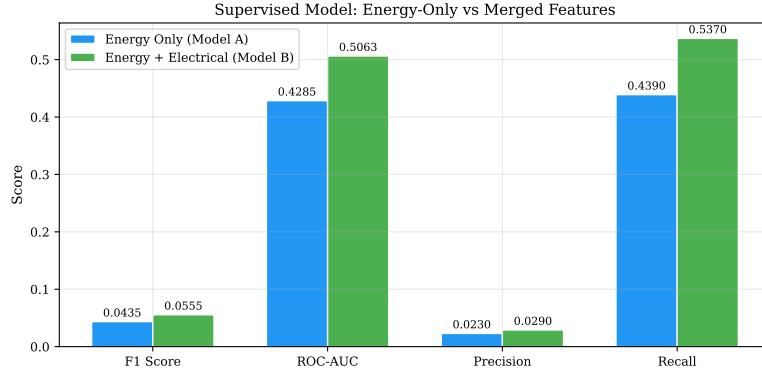


Figure 10: Model performance with energy-only features (Model A) vs. merged energy + electrical features (Model B). The merged model outperforms on all four metrics.